TASK 1

Real Estate Market Data Cleaning

Growfinix Data Analytics Internship — Month 1

Tools: Microsoft Excel, Power Query, VLOOKUP/XLOOKUP

Goal: Clean a messy property-listings dataset, fix missing or inconsistent values, remove duplicates, standardize text, and combine it with historical sales records.

What this guide includes

1. Task objective and workflow
2. Example raw dataset
3. Cleaning steps in Excel / Power Query
4. Lookup formulas to merge sales records
5. Quality checks and final deliverables

Note: The sample data in this guide is fictional and is provided for practice. Replace it with the dataset supplied by your internship coordinator if one was provided.

1. Task Objective

Real estate data often contains repeated property listings, missing ZIP codes, inconsistent spelling, extra spaces, and mismatched property IDs. The aim is to prepare a reliable dataset for analysis.

Expected workflow

Raw listings → Remove duplicates → Handle missing ZIP codes → Standardize text → Match historical sales → Validate → Export clean dataset

2. Example Raw Data

Use the following fictional records to practise. The historical sales table is a separate table to join by Property_ID.

Property_ID	Address	City	ZIP_Code	Property_Type	List_Price
P101	12 lake road	Warangal	506001	House	4500000
P102	7 Green Street	Warangal		Apartment	3200000
P103	7 GREEN STREET	Warangal	506002	apartment	3200000
P104	9 Market Rd	Hanamkonda	506001	House	5100000
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Issues to identify: extra spaces in Address, inconsistent capitalization in Property_Type, a missing ZIP_Code, and a duplicated P104 row. P102 and P103 may look similar, but do not delete records solely because they look similar; confirm using the business rules and unique ID.

Example Historical Sales table

Property_ID	Sale_Date	Sale_Price
P101	2026-01-12	4400000
P102	2026-02-08	3150000
P104	2026-03-20	5000000

3. Clean the Data in Excel

A. Make a working copy

Keep the original file unchanged. Convert the data range into an Excel Table using Ctrl + T. Give it a clear name such as PropertyListings.

B. Remove exact duplicate rows

Select the table → Data → Remove Duplicates. Select the columns that define an exact duplicate. If Property_ID should be unique, investigate repeated IDs before removing anything.

C. Standardize text

Use TRIM to remove extra spaces and PROPER or UPPER when appropriate. Example formulas (assuming Address is in B2 and Property_Type is in E2):

```
=TRIM(B2)
=PROPER(TRIM(E2))
```

Review the result before replacing the original values. PROPER is not suitable for every name or abbreviation.

D. Handle missing ZIP codes

Do not guess a ZIP code. Check a trusted source or a verified address table. If it cannot be confirmed, keep it blank or mark it as “Unknown” according to the project rules, and record how many values remain missing.

4. Clean the Data with Power Query

1. Select a cell in the table and choose Data → From Table/Range.
2. In Power Query, select text columns and choose Transform → Format → Trim, then Clean if needed.
3. Set correct data types: ZIP_Code as Text; List_Price and Sale_Price as Decimal Number or Whole Number; Sale_Date as Date.
4. Remove exact duplicate rows using Home → Remove Rows → Remove Duplicates.
5. Filter blank ZIP codes and resolve them using verified records; do not invent values.
6. Choose Home → Close & Load to return the cleaned table to Excel.

Tip: Keep ZIP codes as Text so leading zeroes are not lost.

5. Merge Historical Sales with XLOOKUP

Assume the cleaned listings table contains Property_ID in column A, and the sales table has Property_ID in column A and Sale_Price in column C on a sheet named Sales.

Use this formula in a new Sale_Price column:

```
=XLOOKUP(A2,Sales!A:A,Sales!C:C,"Not found")
```

This searches for the Property_ID from A2 in the Sales sheet and returns its Sale_Price. If the ID is not found, it displays “Not found”.

Alternative: VLOOKUP

```
=IFERROR(VLOOKUP(A2,Sales!A:C,3,FALSE),"Not found")
```

VLOOKUP requires the lookup ID to be in the first column of the selected range. Check that each Property_ID has the expected number of sales records; if properties can sell more than once, a simple lookup returns only one match.

Power Query merge alternative

In Power Query, select the listings query → Home → Merge Queries → select the sales query → choose Property_ID in both tables → choose Left Outer join → expand the needed sales columns → Close & Load.

6. Validate the Final Dataset

Before submitting, check:

- Duplicate rows were reviewed and removed where appropriate.
- ZIP codes are valid and missing values are documented.
- Text fields have consistent formatting.
- Property_ID values match the sales table correctly.
- Price columns contain numeric values, not text.
- Row counts before and after cleaning are recorded.
- The original data is preserved separately.

7. Suggested Deliverables

1. Original raw dataset.
2. Cleaned Excel workbook (.xlsx).
3. Power Query steps saved in the workbook, if used.
4. Short report describing problems found, cleaning actions, rows removed, missing ZIP codes resolved/unresolved, and lookup/merge results.
5. Screenshots of the cleaning process for your internship submission.

8. Short Conclusion (sample)

“In this task, I cleaned a real estate listings dataset using Excel and Power Query. I checked duplicate records, standardized text, reviewed missing ZIP codes, and merged historical sales information using Property_ID with XLOOKUP/VLOOKUP or Power Query. I validated the final data to make it more consistent and ready for analysis.”

Submission reminder: Use the official internship dataset and follow the instructions from your coordinator. This guide and its sample records are for learning; they are not proof that the task has been completed.